

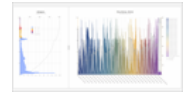
Data Analysis Documentation (Version 17. Onwards)

The Data Analysis Tool is a html+javascript based utility used by administrators, tuning experts, support engineers and consultants to create professional graphs based various types of input data. The tool is fully automatic and significantly faster than Excel at creating scatterplots, bar charts and line plots together with histogram analysis for almost any column separated input data such as **NMON** (*CPU_ALL, MEM, MEM_NEW, PROC, LPAR, PAGE*), **VMSTAT**, **SDFMON**, **TAANA**, **ABAPMETER**, **MHTML**, **NIPING** and **XLSX**. DataAnalysis.HTML will run in all modern web browsers like Chrome, Edge, Firefox, Opera, Safari (*Windows / Linux / MacOS / iPadOS*).

SAP internal note [3169320 - Data Analysis](#)
SAP JAM <https://community.sap.com/t5/technology-blog-posts-by-sap/data-analysis-tool/ba-p/13548321>
Github <https://github.com/i813812/DA-Data-Analysis>

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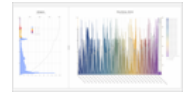


1 Select Input Data

DA Data Analysis allows to open and concatenate multiple input files, it will detect automatically the column separator (*tab, semicolon, comma..*) - any character not included in 0..9, A..Z, a..z or *space* will be detected. To load data into the Data Analysis tool you have three options:

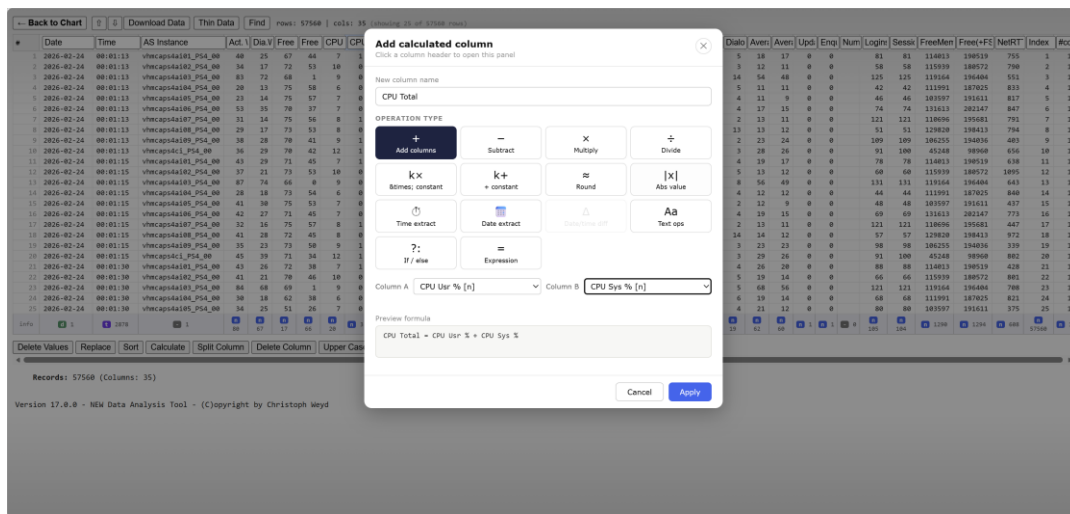
- Use the "Choose Files" button and select one or multiple files (*press **Ctrl** Key to toggle between file and folder selection - if folder is active DA will select all files in a folder*)
- Drag and drop one or multiple files over the "Choose files" button
- Copy & paste any column separated data into the input box
- In some cases when many large numbers with thousand delimiter are included a wrong column separator, typically , or . might be selected - in that case we need to specify the separator manually (*use **S** for space, **T** for tab or specify any other character /|\\.,;-)*)
- **Specify Decimal format** (*point or comma - if numbers are too small the program cannot determine the correct format automatically*)

Paste content of SDFMON into text area – then use button **Update**

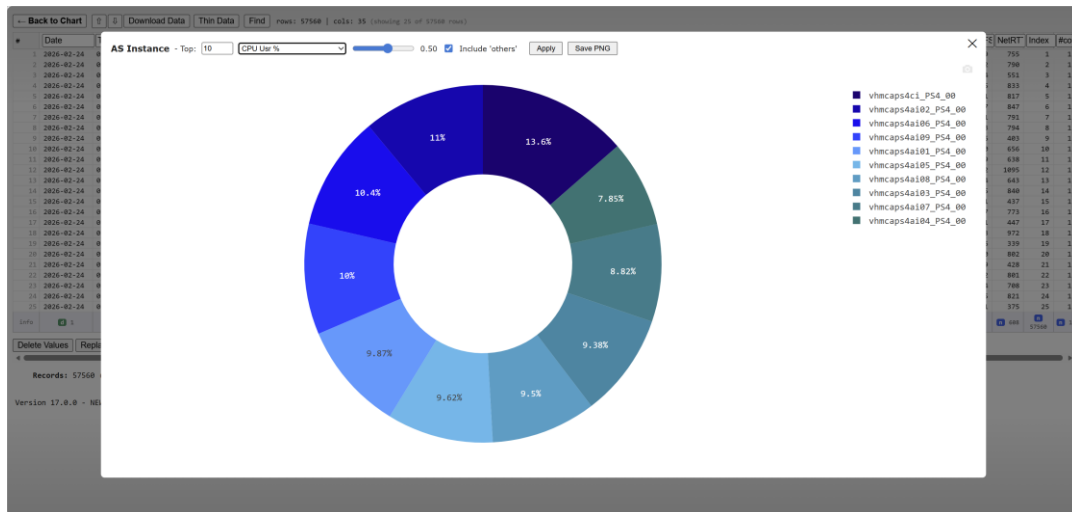
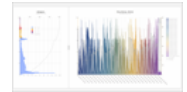


The user has some options to manipulate the input data (*the buttons are only available if the user is putting the cursor into the header line of a column*)

- Find and/or Replace values
- Sort Data
- Delete Values
- Replace Values
- Sort / Rename
- Calculate (create new columns with basic calculations bade on other columns as reference)



- Split Columns
- Delete Columns
- Upper/Lower Case
- Pie Chart (only for non-numeric columns)



On some occasions the program might not be able to determine the date format automatically, in that case the user can choose from different date formats

YYYY-MM-DD

DD-MM-YYYY

YY-MM-DD

DD-MM-YY

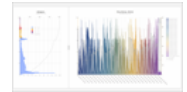
YYYY-DD-MM

MM-DD-YYYY

YY-DD-MM

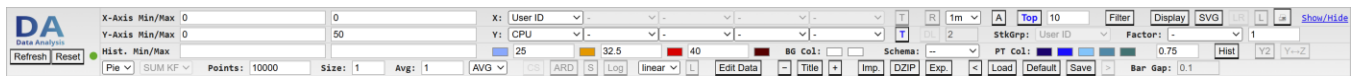
MM-DD-YY

Additionally, the user can download the data (tab separated text file with extension .xls). To reduce the amount of data, columns which do not contain significant information (all values identical) can be removed. If files are too big then the user can also remove every 2nd data row.



3 Customize Graphic Output

Below the various options to customize the graphic output

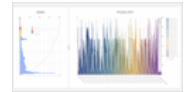


1. **Specify the fields for the X-Axis. Available are non-numeric fields (*date, time, server names...*) and all numerical fields**
2. **Specify the fields for the Y-Axis. Available are only numerical fields (*integer or float*)**

Important: Certain fields for the X-Axis like Date/Time are automatically proposed - for the Y Axis the DataAnalysis will default CPU User, CPU System, Resp.Time...

Optional

3. Activate 2nd Y-Axis and specify fields, chart type...
4. Select the chart type - available options are:
 - Scatter Plot
 - Line Chart
 - Bar Chart
 - Area Chart
5. If multiple key figures are select for the Y-Axis specify if those values should be summarized, stacked over each other, or shown independently
 - Summarize Key Figures
 - Stack Key Figures
 - Independent Key Figures
6. Select if any key figures should be used to group/stack the data. Each different value for the selected Stack variable will be shown in a different color. This option is only available if there are multiple different values available. A graph can only be stacked with this option if for each x-value there are an equal number of y-values available. If any x-value will have a different number of y-values this x-value will not be shown (*this might happen if x-values (time) differ by a few seconds - as workaround one can try to activate rounding*).



4 Additional Options

In the upper left you see the actual status of the Chart and you can Restart, Refresh or Reset the Graph

- **Restart**

If you click on the Logo itself you can restart DA Data Analysis to load another data source.

- Recalculate and Refresh the Graphic

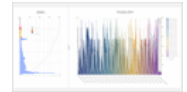
Refresh

- Reset Graphic and recalculate Axis scaling

RS

- **Status**

Ready ● Refresh Pending ● Busy ●



4.1 1st line - X-Axis

X-Axis Min/Max vhmcpa4ai01_PS4_00 00:01:13 vhmcpa4ci_PS4_00 23:59:15 X: AS Instance Time - - - T R 1m A 10 Filter Display SVG L R Show/Hide

- X-Axis Min/Max – Specify the X-Axis Range

X-Axis Min/Max 00:34:00 I64940 23:59:30 I00000

- Select fields for X-Axis (Show/Hide Data Labels)

X: Host Date Time Index

- Show/Hide X-Axis Labels (blue = Show, gray = hide)

T T

- Round time figures (select R to activate) to multiple of 1, 5, 10, 15 ... 60 ... 3600 seconds

R 1m

Available Rounding Values are:

1s
5s
10s
15s
20s
30s
1m
2m
5m
10m
15m
20m
30m
1h

- Specify if data points not present in all stack groups should be added or eliminated

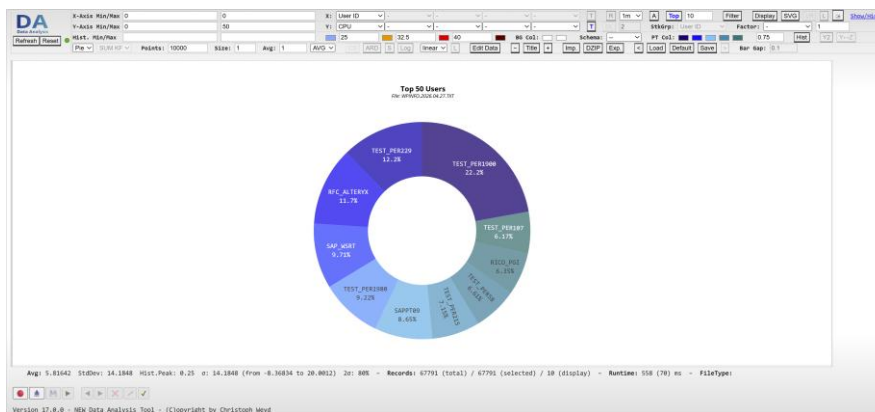
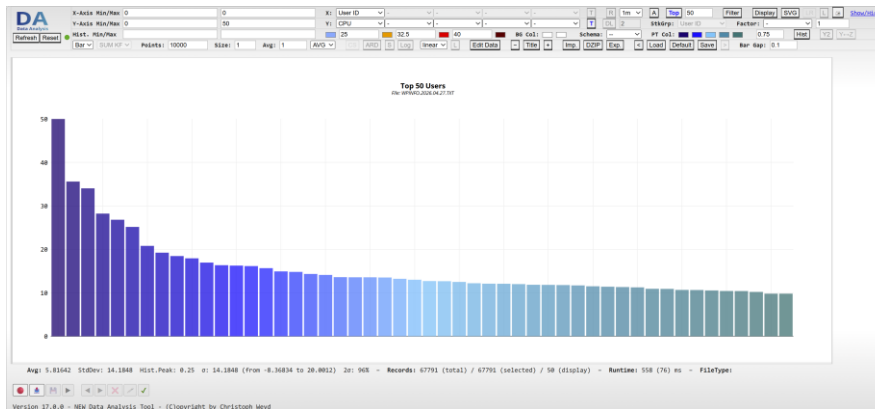
I A E

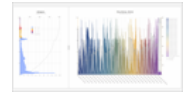
I Ignore data points which are not present in all stack groups
A Add missing data points with zero value
E Erase Data point from all stack groups if not present in all of them

- Top nn (only available for single non-numeric field on X-Axis)

Top 10

Create Bar charts or Pie charts for the Top nn (sorted descending) elements of the select X-Axis with respect to the metric selected for the y-Axis





- Define complex Filter criteria

Filter

multiple conditions can be combined with AND/OR, available operators are

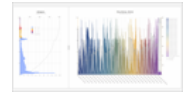
- =
- ≠
- >
- ≥
- <
- ≤
- between
- in
- not in
- contains
- does not contain
- starts with
- ends with
- is empty
- is not empty

- Display Data used in Chart

Display

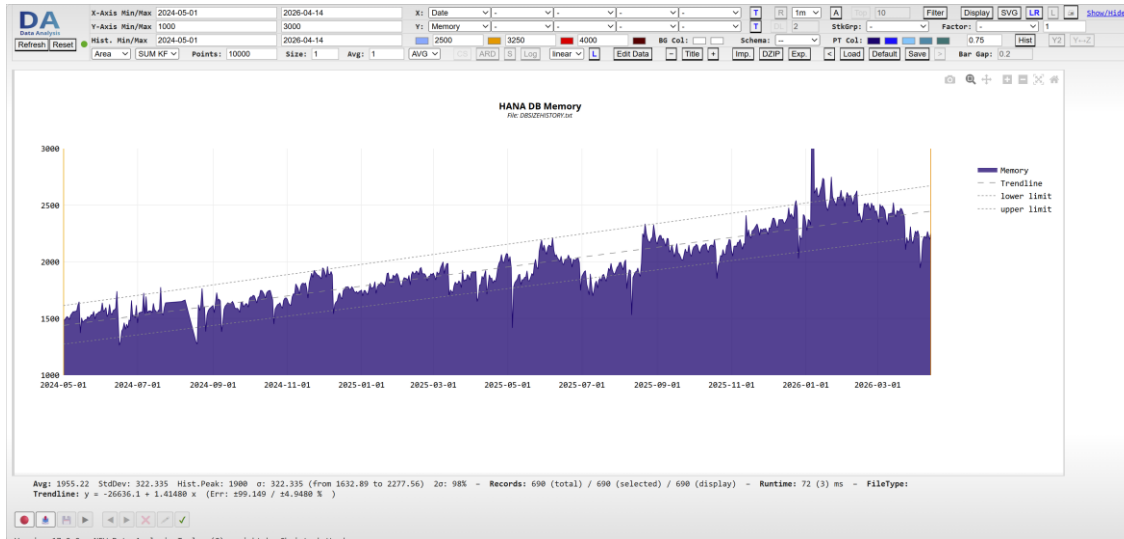
- Save Chart as SVG Vector graphic

SVG



- Create linear regression based trendline (only if X-Axis is Date/Time)

LR



The trendline will indicate the forecast and upper/lower limit; the equation is shown below the graphic. The factor is always measured in xxx per day.

Trendline: $y = -26636.1 + 1.41480 x$ (Err: $\pm 99.149 / \pm 4.9480 \%$)

- Logarithm of Y-Values

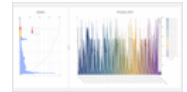
L

Linear

L

Logarithm of Y-Values

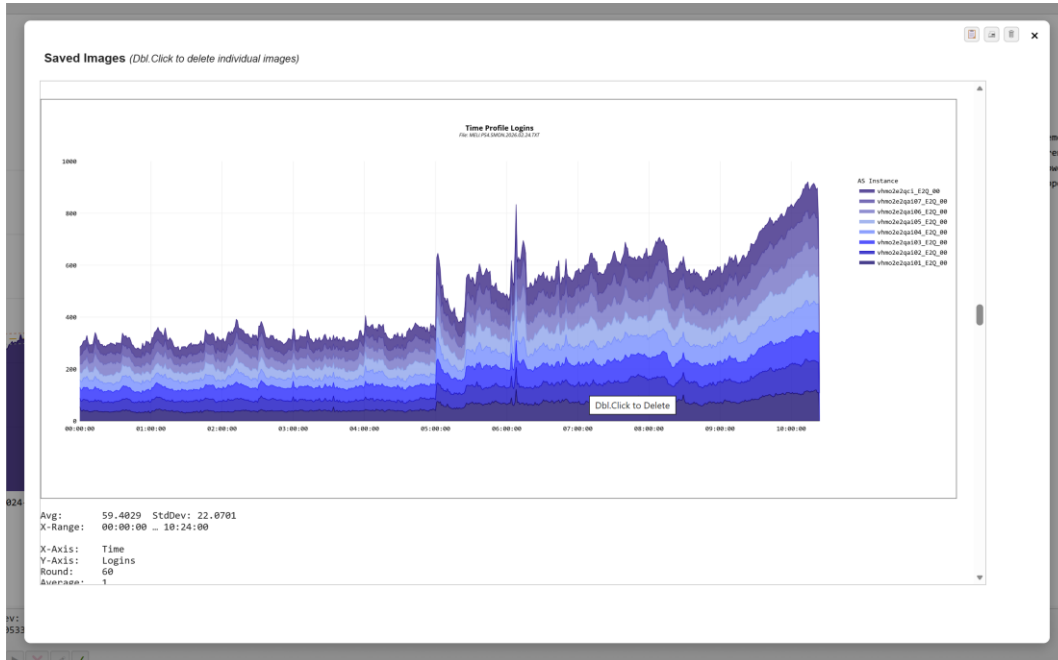
•

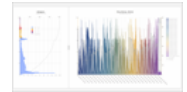


- Print (take screenshot)



This function will take a high-resolution 4k screenshot of the graphic (4500 point horizontal resolution) even on computers which have a low-resolution display. The most recent screenshots are displayed in a modal popup which allows to copy or print all screenshots.





4.2 2nd line - Y-Axis

Y-Axis Min/Max: 0 100 Y: CPU User CPU Sys - - - - T DL 1 StkGrp: AS Instance Factor: - 1

- Y-Axis Min/Max – Specify the Y-Axis Range

Y-Axis Min/Max: 0 50

- Select fields for Y-Axis (Show/Hide Data Labels)

Y: CPU User% CPU Sys% - -

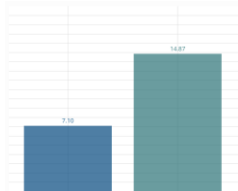
- Show/Hide Y-Axis Labels (blue = Show, gray = hide)

T T

- Data Labels (Bar Charts only)

DL 2

Show data Labels above Bars with n decimals



- Stack Group

StkGrp: Host

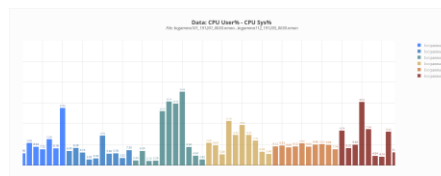
Data Points can be grouped by non-numeric fields as a separate data set, Each Stack Group will show in a separate color which will be interpolated from the four colors defined in the 4th line

Example

X-Axis: Host + Time

Y-Axis: CPU Usage; Stack Group: Host

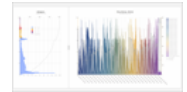
Colors:



- Factor – we can multiply the Y-Values with the value of another column.

Factor: LPAR logicalC 0.01

A useful example is to multiply the CPU% with the total number of logical CPUs to display the logical CPUs used. We add here an additional factor of 0.01 to adjust for the percentage itself.



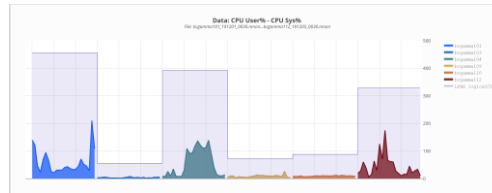
Example

X-Axis: Host + Time

Y-Axis: CPU Usage; Stack Group: Host

Factor: logical CPUs * 0.01

2nd Y-Axis: logical CPUs



4.3 3rd line - 2nd Y-Axis

2nd Y	Min/Max	0	60000000	Z:	Free Mem.	-	-	-	-	T	Area	linear	Avg:	5	Size:	1	Col:	0.25
-------	---------	---	----------	----	-----------	---	---	---	---	---	------	--------	------	---	-------	---	------	------

- 2nd Y-Axis Min/Max – Specify the Y-Axis Range

2nd Y	Min/Max	0	500
-------	---------	---	-----

- Select fields for 2nd Y-Axis (Show/Hide Data Labels)

Z:	LPAR logicalC	-	-	-	-
----	---------------	---	---	---	---

- Show/Hide 2nd Y-Axis Labels (blue = Show, gray = hide)

T	T
---	---

- Details for 2nd Y-Axis

Area	steps	Avg:	1	Size:	1	Col:	0.1
------	-------	------	---	-------	---	------	-----

Here we specify how the 2nd Y-Axis is drawn. From left to right

- Chart Type (Area, Scatter, Line)
- How to connect Data Points (line, spline, steps)
- Average n data points
- Point/Line Size
- Point/Line Color
- Transparency (from 0.05 to 1.00)

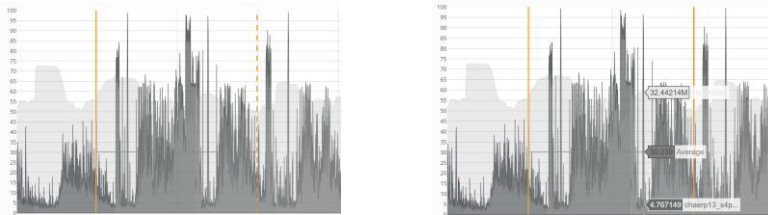
4.4 4th line – Histogram and Colors

Hist. Min/Max chaerp05_S4P_15 C chqreccapp00_S4P_ 65 80 90 BG Col: Schema: Magma PT Col: 0.75 Histogram Display 2nd Y-Axis

- Histogram Range (Display only)

Hist. Min/Max chaerp05_S4P_15 C chqreccapp00_S4P_

Single click on the Chart to set first the left border (the right border will show in a dashed orange line), then click again to set the left border.



- Specify data range and color for the histogram (3rd line)

65 80 90

- Select background and chart color

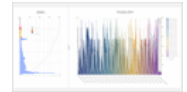
BG Col:

- Color Schema

Schema: Magma

available color options are

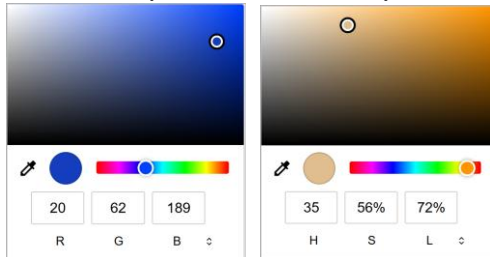
Blue					
Orange					
Red					
Green					
Cyan					
Purple					
Black to White					
Vivid					
Cividis					
Magma					
Spectral					
Rainbow					



- Specify color for the 1st, 2nd, 3rd and 4th select Y-Value and Transparency. If a stack/group variable is used this for color will specify the color range used.



Click on any color to manually select or choose the color picker 



The transparency ranges from 0.05 (almost invisible) to 1.00 (opaque)

- Show/Hide the histogram

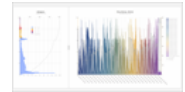
Histogram

- Enable/Disable 2nd Y-Axis

Y2

- Swap primary with 2nd Y-Axis

Y↔Z



4.5 5th line - Chart Options

Area ▾ Sum. Key Figures ▾ Points: 10000 Size: 1 Average: 1 AVG ▾ CS ARD S Lin linear ▾ L Edit Data - Title + Imp. DZIP Exp. < Load Default Save > SVG LR hide

- Select Chart Type - available Chart Types are:

Bar
Scatter
Line
Area

- Handling of Data Points. If multiple key figures are select for the Y-Axis specify if those values should be summarized, stacked over each other or shown independently

Sum. Key Figures
Stack Key Figures
Indep. Key Figures

- Maximum number of points to be drawn - if the data contains after aggregation m points and only x point should be drawn then only every $n^{\text{th}} = m/x$ points will be drawn

Points: 800

- Specify point size (*also line width in points, default = 3*)

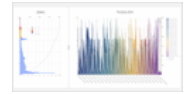
Size: 1

- Select how many data points should be aggregated (*calculate average of n points*)

Average: 1

- Specify calculation type if multiple data points are cumulated - default is AVG (average), available options are:

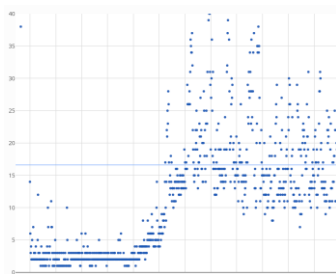
SUM
AVG
MAX
MIN



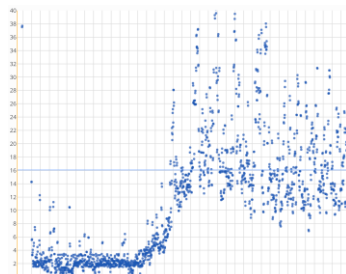
- Add random decimals to prevent banding.

ARD

If the measurements like CPU usage is only available with limited decimals values the graph will look unnatural because all values are on discrete lines. With button S one can add to each key figure a random value r between $-d/2 < r < d/2$ where d is the smallest detected difference between the y values. See below Example



CPU values from SDFMON do not have decimals



Random decimals added (here between ± 0.5)

- Reduce Number of Data Points (Algorithm)

S **T**

S Skip every n 'th data point

T Largest-Triangle-Three-Buckets

(see sveinn-steinarsson/flot-downsample: Downsample plugin for Flot charts. (github.com))

- Switch between linear or logarithmic Y-Axis

Log

Lin

- Specify how data points get connected in a line chart (linear, spline, or steps)

linear

spline

steps

- Display or Hide Chart Legend

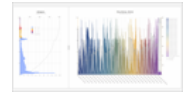
L

- Edit/Preview Data

Edit Data

- Set Chart & Axis Title (supported are the modifiers: ** Bold **, *<i> italic </i>*, _{_{subscript}}, ^{^{superscript}}, and line breaks **
**) and increase or decrease the font size

- **Title** **+**



- Import/export Layout Profile or Export Chart

The two buttons IMP. and Exp. allow to import or export a JSON which contains all chart settings. Those settings can be used when another data source is loaded.

The button DZIP is generating a compressed JSON file which contains the chart settings and the data itself. Typically the DZIP file is around 20% of the original file size.

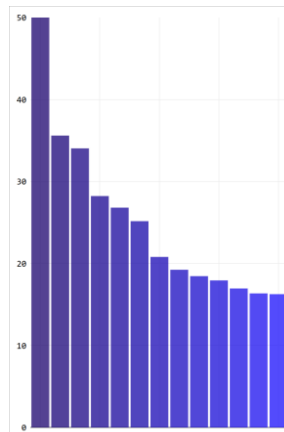
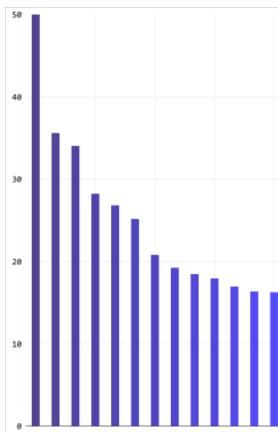
- Load/Save Settings set defaults. Undo/Redo last changes.

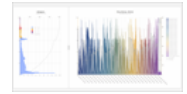
The buttons Load and Save import/export the current chart settings into memory. They can be applied to the next file loaded. The two buttons < > allow to undo or redo the last changes.

- Bar Gap

Bar Gap:

Adjust the size of the gap for Bar charts





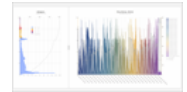
- Logarithm of Y-Values



Linear



Logarithm of Y-Values

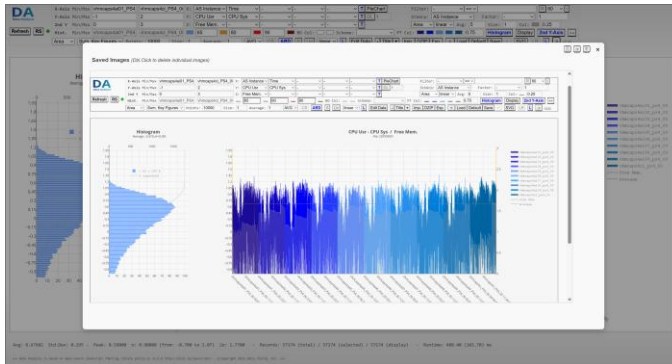


- Store Images as high-resolution PNG graphic with a width of 4000 pixels (600 DPI). Even on low-resolution monitors the program creates a high-res image.



Store and display screenshots

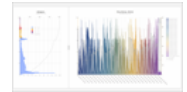
The last 30 previously captured images will be shown in a modal popup



From here one can copy all images to the Clipboard 

- Autohide the chart options (that will make the charts bigger if the cursor is moved over them, the menu is shown again if the cursor is moved to the top)

Hide Show



5 Plotly Options

The below options are visible when you hover the mouse above the graph



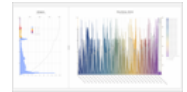
- **Zoom** - use the mouse to select a rectangular area to be shown
- **Move** or **Re-Scale** axis (*move mouse to the end of the axis scales until an arrow is shown, then lets-click and move*)
- **Download** Graph as PNG file
- **Autoscale** / Reset Scale
- **Export** and open chart in Plotly Chart Studio
- Change scaling of X-Axis (double click on graph) – A popup allow to enter the tick distance (in hours, minutes, seconds...)

This page says

Enter tick distance:

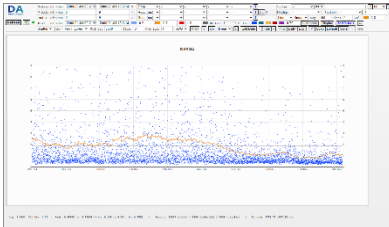
OK Cancel

- If multiple values are shown, click on the legend to show/hide individual key figure or stack values

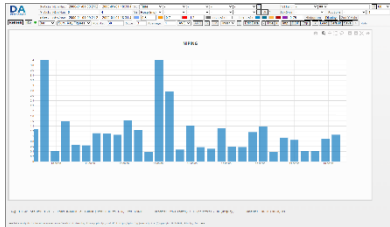


6 Examples

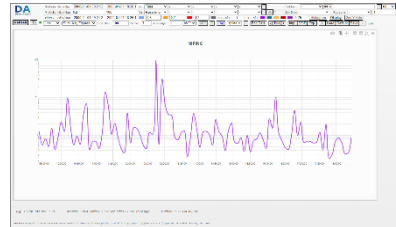
Available graph options are:



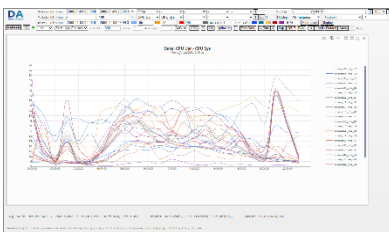
Scatter plot



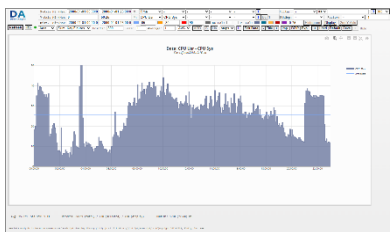
Bar chart



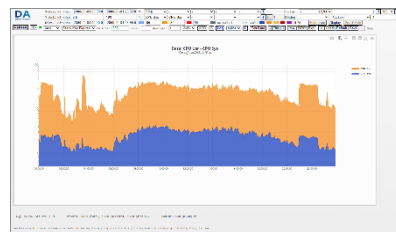
Line chart



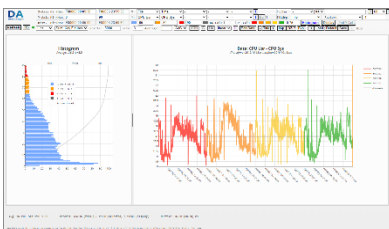
Multi Line Charts



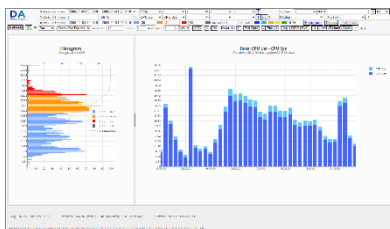
Area Charts



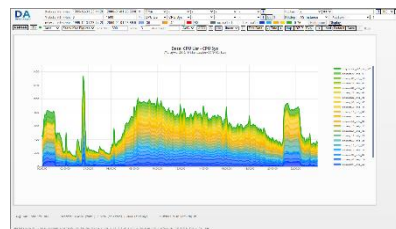
Stacked Area Charts



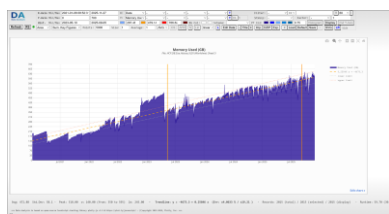
*Grouped Line Charts
(SDFMON per Date)*



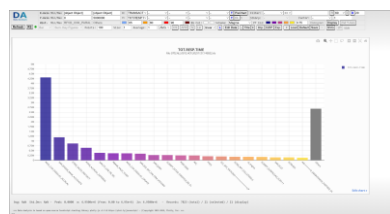
*Stacked Bar Charts
(SDFMON per Appl. Server)*



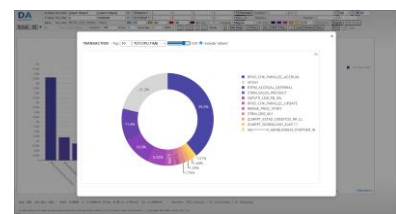
*Stacked Area Charts
(SDFMON per Appl. Server)*



Trendline



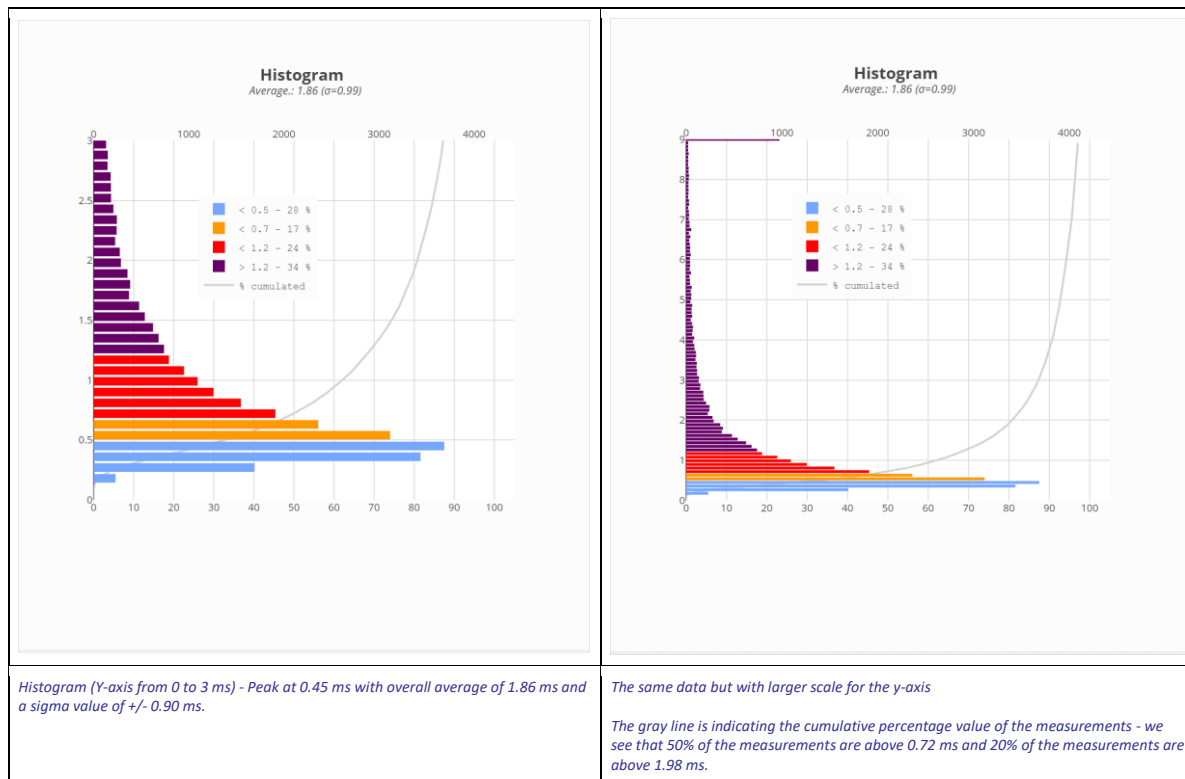
*Top n (descending)
for Bar Charts only*



*Top n Pie Chart
(only for non-numeric fields)*

7 Histogram

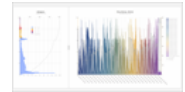
The histogram is calculated for the specified y-axis range divided in 100 equal intervals. For each of those intervals we count how many measurements are recorded and display the result as a horizontal bar chart. The color coding can be specified in the top of the screen:



Especially for highly fluctuating data (*NIPING results, CPU usage, number of active work processes*) a line/bar or scatterplot alone are difficult to interpret and often lead to wrong conclusions. A quantifiable metric to describe the difference in fluctuations is not available on those charts.

Only a histogram analysis allows to exactly describe the random data distribution. Especially for sizing purposes a statement like "95% (2 sigma) of all measurements are below 75 of CPU usage" is much more accurate than the standard deviation a line drawn above the highest observed values. Scatterplots combined with a histogram show the reality, how the individual measurements are fluctuating and at the same time to quantify the fluctuations, distribution and upper lower limits in percentage to the number of recorded measurements. Using maximum/minimum or averages do **NOT** show the reality and should **NOT** be used in the context of capacity analysis. Assuming the histogram follows a gauss normal distribution then we can conclude that:

- In the interval of the deviation $\pm\sigma$ of the expected value, 68.27 % of all measured values can be found,
- In the interval of the deviation $\pm2\sigma$ of the expected value 95.45 % of all measured values are found,



8 Performance

Depending on the file type, record length, number of columns the Data Analysis tool can load and parse approximately **140.000** records per second. The throughput of the graphic engine depends on the number of operations and fields but typically Data Analysis can calculate and draw around **15.000** measurement points per second. We can estimate the average total processing time **T** of the Data Analysis tool by $T(ms) = R/1000 * 7 + P/1000 * 60$ where **R** is the total number of records loaded, **P** the number of data points in the graphic.

Example(s):

- We process 5 NIPING files (*with 1 sec frequency*) which contain a total of **432.000 records** - from those records we plot **20.000 data points**. The estimated total processing time will be at $432 * 5 + 20 * 60 = 4224 ms$ (**4.2 seconds**).
- A single NIPING file of 86.400 records with 10k data point will be ready in $0.60 + 0.60 = 1.2$ seconds.
- We open 30 nmon files (*24h capture, snapshot very 5min, 57 CPU cores, each file with 150.000 records*) with the nmon analyzer to extract CPU_ALL, PROC, LPAR, MEM, MEM_NEW then each file takes about 1 minute of processing time. If we merge all files together with the nmon analyzer the process takes around 30 minutes. The Data Analysis tool can load and display the same data in less than 10 seconds.

9 Memory Usage:

The memory usage of Data Analysis tool depends on the file size and number of data points to be displayed. When we open a SDFMON file with 1.586.920 records the initial memory consumption reached 5GB of memory. During the creation of the graphic the total memory consumption of the browser (opera) increased to 11.5 GB of memory drawing all 1.586.920 data points (runtime 75 seconds). When we limit the number of data points to 200.000 the memory consumption reached only 2.1GB. The memory consumption scales with the number of data points drawn in the graphic and can be estimated at about 5-10 kB per data point.

10 Errors / Bugs

In case of any errors, please send an email to c.weyd@sap.com and include a copy (*or link*) to the used data source and a screenshot of the browser window (*including the used settings in the top*). Please also describe which browser and operating system was used.

Data Analysis is based on open-source JavaScript charting library [plotly.js](https://plot.ly/javascript/) <https://plot.ly/javascript/>

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